

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE273	FOUNDATIONS OF MACHINE LEARNING	2	0	2	3	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50					

Course Outcomes :Through this course students should be able to

CO1 :: Explain the fundamentals of Machine Learning, its types, and differences from AI and Data Science.

CO2 :: Apply Python libraries (Pandas, Matplotlib) to load, inspect, manipulate, and visualize datasets for meaningful insights.

CO3 :: Apply exploratory data analysis techniques (univariate and bivariate plots, correlation analysis) to identify basic patterns and trends in datasets.

CO4 :: Apply linear algebra, probability, and calculus concepts to interpret the mathematical foundations of machine learning algorithms.

CO5 :: Apply statistical techniques and optimization methods to analyze data and understand model training and convergence behavior in machine learning.

CO6 :: Understand how machine learning models learn, generalize, and are chosen based on data

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	MACHINE LEARNING USING PYTHON	MANARANJAN PRADHAN, U DINESH KUMAR	WILEY
	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	MACHINE LEARNING IN DATA SCIENCE USING PYTHON	DR. R. NAGESWARA RAO	WILEY
R-2	REAL-WORLD MACHINE LEARNING	HENRIK BRINK, JOSEPH W. RICHARDS, MARK-FETHEROLF, BEAU CRONIN	WILEY
R-3	MACHINE LEARNING	ANURADHA SRINIVASARAGHAVAN , VINCY JOSEPH	WILEY
R-4	DATA SCIENCE AND MACHINE LEARNING USING PYTHON	REEMA THAREJA	MC GRAW HILL

R-5	MATHEMATICS FOR MACHINE LEARNING	MARC PETER DEISENROTH, A. ALDO FAISAL, CHENG SOON ONG	CAMBRIDGE UNIVERSITY PRESS
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LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Machine Learning Foundation(Introduction to Machine Learning, Types of Machine Learning, Difference between ML, AI and Data Science)	T-1		L1: Lecture-0 to be delivered arousing students' interest in the course and its importance. L2: Introduction to Machine Learning, Types of Machine Learning, Difference between ML, AI and Data Science	Students would be able to understand the course details, significance of the course and introduction of Machine Learning and its types.	Presentation and Discussion	
	Lecture 2	Machine Learning Foundation(Introduction to Machine Learning, Types of Machine Learning, Difference between ML, AI and Data Science)	T-1		L1: Lecture-0 to be delivered arousing students' interest in the course and its importance. L2: Introduction to Machine Learning, Types of Machine Learning, Difference between ML, AI and Data Science	Students would be able to understand the course details, significance of the course and introduction of Machine Learning and its types.	Presentation and Discussion	

Week 2	Lecture 3	Machine Learning Foundation(Parametric vs Non-parametric models, Generative vs Discriminative Models, Machine Learning Workflow)	T-1		Parametric vs Non-parametric models, Generative vs Discriminative Models, Machine Learning Workflow	Students would be able to understand various machine learning models and the machine learning workflow.	Presentation and Discussion	
	Lecture 4	Machine Learning Foundation(Train-Test data split, Overview of Evaluation Metrics)	T-1		Train-Test data split, Overview of Evaluation Metrics	Understand how to split data into training set, test set and validation set; and understand various Evaluation metrics for Machine learning model	Programming hands-on using Python	
Week 3	Lecture 5	Machine Learning Foundation(Overfitting and Underfitting, Bias-Variance Trade-off)	T-1		Overfitting and Underfitting, Bias-Variance Trade-off	Understand the concepts and implications of Overfitting and Underfitting, Bias-Variance Trade-off.	Presentation and Discussion	
		Machine Learning Foundation(Applications and Use-cases of ML)	T-1		Applications and Use-cases of ML	Understand ML applications and use-cases in various fields.	Presentation and Discussion	
	Lecture 6	Numpy for Numerical Computing(Creating Arrays, Operations on Arrays, Broadcasting Rule, Matrix operations, Random Module)	T-1 R-1 R-4		Creating Arrays, Operations on Arrays, Broadcasting Rule, Matrix operations, Random Module	Understand Numpy arrays and related operations and Random module of Python	Programming hands-on using Python	
Week 4	Lecture 7	Pandas for Data Handling (Series and Data Frame, Reading and writing CSV file)	T-1 R-1 R-4		Pandas Library - Series and Data Frame, Reading and writing CSV file	understand Pandas Library and reading writing CSV files in Python.	Programming hands-on using Python	
		Pandas for Data Handling (Inspecting Datasets, Selecting and filtering data)	T-1 R-1 R-4		Pandas Library - Inspecting Datasets, Selecting and filtering data	understand Pandas Library and selecting and filtering data.	Programming hands-on using Python	
	Lecture 8	Data Visualization using Matplotlib(Line plot, Bar plot, Histogram)	T-1 R-1 R-4		Matplotlib - Creating Line plot, Bar plot, Histogram	Understand how to create line chart, bar chart, and histograms using Matplotlib.	Programming hands-on using Python	
		Data Visualization using Matplotlib(Scatter plot, Pie chart, Customizing plots, Subplots)	T-1 R-1 R-4		Matplotlib - Creating Scatter plot, Pie chart, Customizing plots, Subplots	Understand how to create scatter charts, pie charts and multi-chart plots using Matplotlib.	Programming hands-on using Python	

Week 5	Lecture 9	Exploratory Data Analysis (Types of data, Loading data using Pandas)	T-1 R-1 R-4		Types of data, Loading data using Pandas	Understand types of data and how to load data using Pandas.	Programming hands-on using Python	
		Exploratory Data Analysis (Univariate analysis using Histogram, Box plot, Count plots)	T-1 R-1 R-4		Univariate analysis using Histogram, Box plot, Count plots	Understand how to perform univariate analysis using histograms, box plots and count plots.	Programming hands-on using Python	
	Lecture 10	Exploratory Data Analysis (Bivariate analysis using Scatter plots, Line plots)	T-1 R-1 R-4		Bivariate analysis using Scatter plots, Line plots	Understand how to perform Bivariate analysis using Scatter plots and Line plots.	Programming hands-on using Python	
Week 6	Lecture 11	Exploratory Data Analysis (Violin plots for category vs numeric, Correlation analysis using Heatmaps)	T-1 R-1 R-4		Violin plots for category vs numeric, Correlation analysis using Heatmaps	Understand how to use Violin plots for category vs numeric data, and analyze Correlation analysis using Heatmaps.	Programming hands-on using Python	
	Lecture 12	Exploratory Data Analysis (Multicollinearity, Distribution analysis: Skewness, Kurtosis, Detecting patterns, anomalies and trends)	T-1 R-1 R-4		Multicollinearity, Distribution analysis: Skewness, Kurtosis, Detecting patterns, anomalies and trends	Understand Multicollinearity, Distribution analysis and Detecting patterns, anomalies and trends.	Presentation and Discussion	
Week 7	Lecture 13	Exploratory Data Analysis (EDA workflow integration)	T-1 R-2 R-3		EDA workflow integration	Understand a complete integration of a EDA workflow.	Presentation and Discussion	
SPILL OVER								
Week 7	Lecture 14				Spill Over			
MID-TERM								
Week 8	Lecture 15	Linear Algebra and Calculus (Scalars, Vectors, Matrices, Tensors)	T-1 R-5		Scalars, Vectors, Matrices, Tensors	Understand the concepts of Scalars, Vectors, Matrices and Tensors.	Presentation and Discussion	
		Linear Algebra and Calculus (Eigenvalues and Eigenvectors)	T-1 R-5		Scalars, Vectors, Matrices, Tensors	Understand the concepts of Eigenvalues and Eigenvectors.	Presentation and Discussion	
	Lecture 16	Probability Foundations (Random variables, Probability distribution)	T-1 R-5		Random variables, Probability distribution	Understand the concepts of Random Variables and Probability Distributions.	Presentation and Discussion	

Week 8	Lecture 16	Probability Foundations (Mean, Variance, Covariance)	T-1 R-5		Random variables, Probability distribution	Understand the concepts of Mean, Variance, Covariance.	Mathematical Illustrations	
Week 9	Lecture 17	Probability Foundations (Joint, Marginal and Conditional Probability)	T-1 R-5		Joint, Marginal and Conditional Probability	Understand the concepts of Joint, Marginal and Conditional Probability	Mathematical Illustrations	
		Probability Foundations (Baye's Theorem, Likelihood vs Probability)	T-1 R-5		Baye's Theorem, Likelihood vs Probability	Understand the concepts of Baye's Theorem, and difference between Likelihood and Probability.	Presentation and Discussion	
	Lecture 18	Calculus for ML(Functions, Gradient)	T-1 R-5		Functions, Gradient	Understand the concepts of Functions and Gradient.	Presentation and Discussion	
		Calculus for ML(Partial derivatives, Chain Rule)	T-1 R-5		Functions, Gradient	Understand the concepts of Partial derivatives, Chain Rule.	Presentation and Discussion	
Week 10	Lecture 19	Statistics for ML(Sampling, Population, Hypothesis Testing)	T-1		Sampling, Population, Hypothesis Testing	Understand the concepts of Sampling, Population and Hypothesis Testing.	Presentation and Discussion	
	Lecture 20	Statistics for ML (Confidence Intervals, Correlation, Maximum Likelihood Estimation)	T-1		Confidence Intervals, Correlation, Maximum Likelihood Estimation	Understand the concepts of Confidence Intervals, Correlation, and Maximum Likelihood Estimation.	Presentation and Discussion	
Week 11	Lecture 21	Optimization Methods(Loss Functions, Convex vs Non-convex Functions, Learning Rate Impact, Local Minima, Global Minima, Saddle Points)	T-1		Loss Functions, Convex vs Non-convex Functions, Learning Rate Impact, Local Minima, Global Minima, Saddle Points	Understand concepts of loss functions, Learning Rate, Minima and Saddle points.	Presentation and Discussion	
	Lecture 22	Optimization Methods (Gradient Descent, Stochastic Gradient Descent (SGD))	T-1 R-3		Gradient Descent, Stochastic Gradient Descent (SGD)	Understand the concepts of Gradient Descent, and Stochastic Gradient Descent (SGD)	Presentation and Discussion	

Week 12	Lecture 23	Optimization Methods (Advanced optimizers (Momentum, RMSProp, Adam))	T-1		Advanced optimizers (Momentum, RMSProp, Adam)	Understand use of Advanced Optimization techniques	Presentation and Discussion	
	Lecture 24	Statistical Learning Theory (Well-Posed Learning Problems, Components of a Learning System, Statistical Learning Framework, Empirical Risk Minimization)	R-1 R-2 R-3		This lecture introduces the fundamental theoretical concepts that form the backbone of Machine Learning. It begins with an explanation of well-posed learning problems, outlining how learning tasks are formally defined using task, experience, and performance measures. The session then explores the core components of a learning system, including data, feature representation, models, learning algorithms, and evaluation metrics. The lecture further presents the statistical learning framework, empha	Understand the concept of a well-posed learning problem by identifying task, experience, and performance measures and Understand the key components of a machine learning system, including data, feature representation, models, learning algorithms, and evaluation metrics.	Presentation and Discussion	Statistical learning framework: Emails are assumed to be sampled from an unknown distribution; the model learns a function that minimizes expected classification error.

Week 13	Lecture 25	Statistical Learning Theory (Inductive Bias, Role of Prior Knowledge, Probably Approximately Correct Learning)	R-1 R-2 R-3		This lecture introduces the theoretical foundations of machine learning by exploring inductive bias and the role of prior knowledge in enabling models to generalize from limited data, followed by the Probably Approximately Correct (PAC) learning framework that formally defines learnability in terms of accuracy, confidence, and computational feasibility. It further examines sample complexity to understand how the amount of training data influences learning performance and model reliability. The l	Understand the concept of inductive bias and its necessity in enabling machine learning models to generalize beyond training data and Understand the principles of Probably Approximately Correct (PAC) learning and how learnability is defined using accuracy and confidence measures and also Understand how to select appropriate machine learning algorithms based on data characteristics, assumptions, and problem-specific constraints.	Presentation and Discussion	Spam Email Classification:
	Lecture 26	Statistical Learning Theory (Sample Complexity, No Free Lunch Theorem, Choosing Algorithms Based on Data and Assumptions)	R-1 R-2 R-3		This lecture explores Sample Complexity as a measure of how much data is required for a learning algorithm to generalize reliably, introduces the No Free Lunch Theorem to emphasize that no single algorithm performs best for all problems, and explains how choosing algorithms based on data characteristics and underlying assumptions leads to effective model selection. The session connects theoretical guarantees with practical decision-making in machine learning by highlighting the role of data dist	Understand the concept of sample complexity and its importance in determining the amount of training data needed for reliable learning and Understand the implications of the No Free Lunch Theorem on algorithm performance and model selection and also Understand how data properties and assumptions guide the choice of appropriate machine learning algorithms.	Presentation and Discussion	

Week 14	Lecture 27	Statistical Learning Theory (Sample Complexity, No Free Lunch Theorem, Choosing Algorithms Based on Data and Assumptions)	R-1 R-2 R-3		This lecture explores Sample Complexity as a measure of how much data is required for a learning algorithm to generalize reliably, introduces the No Free Lunch Theorem to emphasize that no single algorithm performs best for all problems, and explains how choosing algorithms based on data characteristics and underlying assumptions leads to effective model selection. The session connects theoretical guarantees with practical decision-making in machine learning by highlighting the role of data dist	Understand the concept of sample complexity and its importance in determining the amount of training data needed for reliable learning and Understand the implications of the No Free Lunch Theorem on algorithm performance and model selection and also Understand how data properties and assumptions guide the choice of appropriate machine learning algorithms.	Presentation and Discussion	
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Test - Code based 2	50	CO4, CO5
Test - Code based 1	50	CO1, CO2, CO3
Recoup CA		

Details of Academic Task(s)

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Test - Code based 1	Assess the students understanding of ML basics, Numpy, Pandas, Matplotlib and EDA	MCQ test based on syllabus from Units-1,2,3	Individual	Online	30	4 / 6
Test - Code based 2	Assess the students understanding of Linear Agenra, Calculus, Probability, and Statistics for ML	MCQ test based on syllabus from Units-4,5	Individual	Online	30	10 / 12
Recoup CA	subjective test based on the syllabus	Additional chance of CA	Individual	Online	30	8 / 10

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	List of Practicals	01. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.		Implementing NumPy arrays and their operations
Practical 2	List of Practicals	01. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.		Implementing NumPy arrays and their operations
Practical 3	List of Practicals	01. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.		Implementing NumPy arrays and their operations
Practical 4	List of Practicals	01. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.		Implementing NumPy arrays and their operations

Practical 5	List of Practicals	02. Load a CSV dataset and explore its structure using Pandas, and create appropriate visualizations to understand data distribution and relationships using Matplotlib.		Processing CSV files and data frames using Pandas and visualizing data using Matplotlib
Practical 6	List of Practicals	02. Load a CSV dataset and explore its structure using Pandas, and create appropriate visualizations to understand data distribution and relationships using Matplotlib.		Processing CSV files and data frames using Pandas and visualizing data using Matplotlib
Practical 7	List of Practicals	02. Load a CSV dataset and explore its structure using Pandas, and create appropriate visualizations to understand data distribution and relationships using Matplotlib.		Processing CSV files and data frames using Pandas and visualizing data using Matplotlib
Practical 8	List of Practicals	02. Load a CSV dataset and explore its structure using Pandas, and create appropriate visualizations to understand data distribution and relationships using Matplotlib.		Processing CSV files and data frames using Pandas and visualizing data using Matplotlib
Practical 9	List of Practicals	03. Perform Exploratory Data Analysis on the Titanic dataset to study passenger demographics, survival patterns, distribution of numerical features, and relationships between categorical and numerical variables.		Understanding relationships between variables/features
Practical 10	List of Practicals	03. Perform Exploratory Data Analysis on the Titanic dataset to study passenger demographics, survival patterns, distribution of numerical features, and relationships between categorical and numerical variables.		Understanding relationships between variables/features
Practical 11	Test - Code based 1			
Practical 12	Test - Code based 1			
Practical 13	List of Practicals	04. Conduct EDA on the Students Performance dataset to analyze score distributions, the impact of parental education and study hours on student performance, and correlations among numerical attributes.		Understanding Correlation among numerical attributes

Practical 14	List of Practicals	04. Conduct EDA on the Students Performance dataset to analyze score distributions, the impact of parental education and study hours on student performance, and correlations among numerical attributes.		Understanding Correlation among numerical attributes
Practical 15	List of Practicals	05. Implement vector and matrix operations using NumPy, compute eigenvalues and eigenvectors, and numerically calculate the gradient of a given multivariable function using partial derivatives.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.
	List of Practicals	06. Simulate random variables from different probability distributions and compute mean, variance, covariance, and conditional probability.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.
Practical 16	List of Practicals	06. Simulate random variables from different probability distributions and compute mean, variance, covariance, and conditional probability.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.
	List of Practicals	05. Implement vector and matrix operations using NumPy, compute eigenvalues and eigenvectors, and numerically calculate the gradient of a given multivariable function using partial derivatives.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.
Practical 17	List of Practicals	05. Implement vector and matrix operations using NumPy, compute eigenvalues and eigenvectors, and numerically calculate the gradient of a given multivariable function using partial derivatives.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.
	List of Practicals	06. Simulate random variables from different probability distributions and compute mean, variance, covariance, and conditional probability.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.
Practical 18	List of Practicals	06. Simulate random variables from different probability distributions and compute mean, variance, covariance, and conditional probability.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.
	List of Practicals	05. Implement vector and matrix operations using NumPy, compute eigenvalues and eigenvectors, and numerically calculate the gradient of a given multivariable function using partial derivatives.		Implementing eigen values, gradients, mean, variance, conditional probability simulating random variables.

Practical 19	List of Practicals	07. Using the Students Performance dataset, draw samples to estimate population parameters, compute correlations, construct confidence intervals, and perform a hypothesis test on mean score differences.		Implementing computation of correlation and Hypothesis Test
Practical 20	List of Practicals	07. Using the Students Performance dataset, draw samples to estimate population parameters, compute correlations, construct confidence intervals, and perform a hypothesis test on mean score differences.		Implementing computation of correlation and Hypothesis Test
Practical 21	List of Practicals	08. Using the Boston Housing dataset, implement and compare Gradient Descent and its variants (SGD, Momentum, RMSProp, Adam) on the regression loss function to study learning rate impact and convergence behavior.		Implementing and comparing Gradient Descent and its variants
Practical 22	List of Practicals	08. Using the Boston Housing dataset, implement and compare Gradient Descent and its variants (SGD, Momentum, RMSProp, Adam) on the regression loss function to study learning rate impact and convergence behavior.		Implementing and comparing Gradient Descent and its variants
Practical 23	Test - Code based 2			
Practical 24	Test - Code based 2			
Practical 25	List of Practicals	09. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.		Implementing Data Pre-processing tasks
Practical 26	List of Practicals	09. Write a program using NumPy to create and manipulate arrays, perform vectorized operations, apply broadcasting, and execute basic matrix computations.		Implementing Data Pre-processing tasks
Practical 27	List of Practicals	10. Write a NumPy program to generate a random 3×3 matrix and perform transpose, trace, rank, matrix multiplication with its transpose, and inverse computation (if possible)		Implementing feature engineering

Practical 28	List of Practicals	10. Write a NumPy program to generate a random 3×3 matrix and perform transpose, trace, rank, matrix multiplication with its transpose, and inverse computation (if possible)		Implementing feature engineering
	SPILL OVER			
Practical 29	Spill Over			